

Scheme 2

Stage 2 Abstract-Level Structured Coding Scheme

Corpus-wide abstract coding of BPS usage, domains, and typology

DRAFT FOR EXPERT EVALUATION. These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes in the Python workflow with an LLM (gemini-2.5-flash); the full end-to-end run is deliberately held until after expert feedback, and the refinements proposed here are awaiting sign-off before any re-run.

At a Glance

Workflow position

Abstract coding for every Stage 1 included record, before Stage 3 candidate selection.

Operational mode

Structured LLM coding with archived JSON batches, deterministic vocabulary normalization, and a rule-based fallback when model output is incomplete or unavailable.

Unit of analysis

One included review record, coded from title, abstract, publication types, and journal metadata only.

Provenance basis

The actual Stage 2 output schema in stage2_abstract_coding.csv.

Research questions

RQ1 (BPS operationalization); RQ3 (concepts and frameworks); SQ1 (conceptual problems); Feeds the Stage 3 candidate gate

Tagline

Structured LLM-first coding with deterministic repair and rule-based fallback

Canonical Source Paths

- `src/protocol/codebooks/stage2_codebook.md`
- `src/review_stages/04_extraction/codebooks/stage2_codebook.csv`
- `src/bps_review/extraction/stage2.py`
- `src/bps_review/extraction/llm_stage2.py`
- `src/review_stages/04_extraction/outputs/stage2_abstract_coding.csv`
- `src/review_stages/04_extraction/outputs/stage2_llm_structured_coding.csv`
- `src/review_stages/04_extraction/outputs/llm_stage2_structured_batches.jsonl`

Purpose

This scheme standardizes abstract-level extraction for all eligible chronic pain reviews. It is the main corpus-wide coding layer used to describe review characteristics, classify the function of biopsychosocial language, detect biological, psychological, and social content, flag conceptual problems, and generate a provisional biopsychosocial typology for downstream synthesis.

It also sets the Stage 3 candidate gate: a record coded as musculoskeletal or as unspecified-but-not-excludable is carried forward for full-text work.

Carried-Through Metadata Fields

Descriptive fields retained from earlier stages for synthesis.

- record_id, database, pmid, pmcid, doi
- title, abstract (primary coding text)
- year, journal, authors, country_contact_author
- publication_types (source publication-type metadata)
- objective_text (objective sentence extracted from the abstract)
- screening_status, screening_reason (inherited from Stage 1)

Coded Fields and Controlled Values

The operational Stage 2 vocabulary. Anchors and boundaries below are the proposed reliability layer over the existing value lists.

review_type

Evidence-synthesis design as stated in the abstract or publication metadata.

- **systematic review** Explicit systematic methods (protocol, search, screening).
- **meta-analysis** Quantitative pooling of effect sizes.
- **network meta-analysis** Multiple-treatment comparison with indirect evidence.
- **umbrella review** Review of reviews.
- **scoping or mapping review** Breadth-oriented mapping of a field without pooled effects.
- **rapid review** Streamlined systematic methods.
- **realist review** Theory-driven mechanism review.
- **integrative review** Mixed evidence integration.
- **narrative or expert review** Non-systematic expert synthesis.
- **other evidence synthesis** A synthesis type not captured above.
- **unclear** Design cannot be determined from the abstract.

Boundary: prefer the most specific applicable label; meta-analysis outranks systematic review when pooling is explicit.

objective_category

Primary stated purpose of the review.

- **conceptual** Purpose is definitional, theoretical, or about operationalizing a model. *Choose when:* framework, concept, operationalize, model. *Not this if:* A treatment-effect question, which is clinical.

- **clinical** Purpose concerns treatment, management, rehabilitation, or care. *Choose when:* treatment, intervention, management, rehabilitation.
- **methodological** Purpose concerns measurement, psychometrics, or assessment tools. *Choose when:* measurement, psychometric, instrument, assessment.
- **epidemiological** Purpose concerns prevalence, incidence, risk factors, or associations. *Choose when:* prevalence, incidence, risk factor, association.
- **mixed** Two or more purposes carry comparable weight.
- **unclear** Purpose cannot be determined.

Boundary: choose the dominant purpose; use mixed only when two purposes are genuinely co-primary.

objective_category_source

Whether the objective classification came from the structured LLM, a repaired LLM batch, or the deterministic fallback.

icd11_pain_category

ICD-11 aligned pain category inferred from the abstract.

- **chronic secondary musculoskeletal pain** Low back, neck, osteoarthritis, whiplash, and similar.
- **chronic neuropathic pain** Neuropathic, CRPS, radiculopathy.
- **chronic cancer-related pain** Cancer or oncology pain.
- **chronic postsurgical or posttraumatic pain** Post-surgical or post-traumatic persistent pain.
- **chronic secondary headache or orofacial pain** Headache, migraine, orofacial, TMD.
- **chronic secondary visceral pain** Visceral, abdominal, pelvic pain.
- **chronic primary pain** Fibromyalgia and other primary pain syndromes.
- **mixed or unspecified chronic pain** Chronic pain with no single dominant site.
- **unclear** Category cannot be inferred.

Abstract-based and allowed to remain unclear. Musculoskeletal and mixed-or-unspecified both feed the Stage 3 candidate gate.

musculoskeletal_flag

Whether the review concerns musculoskeletal pain. Routes records into the musculoskeletal review.

- **yes** Musculoskeletal focus is explicit.
- **no** Clearly non-musculoskeletal.
- **unclear** Cannot be determined; carried forward.

neuropathic_flag

Whether the review concerns neuropathic pain. Parallel to `musculoskeletal_flag`; routes records into the neuropathic review. A record may set both flags (for example a mixed review) or neither.

- **yes** Neuropathic focus is explicit (for example neuropathy, radiculopathy, CRPS, postherpetic or diabetic neuropathic pain).
- **no** Clearly non-neuropathic.
- **unclear** Cannot be determined; carried forward.

stage3_track

Which review or reviews the record is routed to. Derived from the two flags; a record can belong to both pools.

- **musculoskeletal** Musculoskeletal review only.
- **neuropathic** Neuropathic review only.
- **both** Enters both pools (mixed or unresolved pain family).
- **none** Neither track; not a Stage 3 candidate.

bps_mention_location

Where the BPS term appears.

- **title only** Only in the title.
- **abstract only** Only in the abstract.
- **title and abstract** In both.
- **unclear** Location cannot be resolved.

bps_function

The rhetorical and analytic work the BPS label performs. This is the central RQ1 field and the finest-grained one.

- **explanatory framework** BPS is used as a model that explains pain or pain-related disability. *Choose when:* explains, accounts for, mechanism of. *Not this if:* Only justifies multimodal treatment, which is intervention rationale. *Boundary:* Requires explanatory intent, not only endorsement of a model's existence.
- **intervention rationale** BPS mainly justifies multimodal or interdisciplinary treatment. *Choose when:* supports multidisciplinary care. *Not this if:* Explains pain aetiology, which is explanatory framework.
- **organizing principle** BPS structures the scope or categories of the review without specifying integration mechanisms. *Choose when:* We organize factors into bio, psycho, social. *Not this if:* A stated cross-domain mechanism.
- **justification** BPS justifies the relevance or importance of the review topic.
- **background framing** BPS sets context in the background without analytic use.
- **conclusion** BPS appears mainly in the concluding claims.
- **policy/practice implication** BPS frames a policy or practice recommendation.
- **rhetorical label** BPS is invoked ceremonially, aspirationally, or symbolically with no substantive analytic work. *Choose when:* a biopsychosocial approach is needed, with no follow-through. *Not this if:* Any substantive explanatory or organizing use.
- **unclear** Function cannot be determined.

bio_mentioned / psych_mentioned / social_mentioned

Binary presence of each domain in the abstract.

- **yes** The domain is present beyond the mere BPS token.
- **no** The domain is not present.

See the proposed graded-mention refinement below; the current field is binary.

quality_assessment_reported

Whether the abstract reports a risk-of-bias or quality assessment.

- **yes** AMSTAR, risk of bias, or quality assessment named.
- **no** Not reported.
- **unclear** Cannot be determined.

psychological_concepts_detected

Normalized list of psychological concepts from title and abstract. Feeds Scheme 5. (*free text*)

theoretical_frameworks_detected

Normalized list of named frameworks or model labels. (*free text*)

conceptual_problem_flags

Provisional abstract-level conceptual problems.

- **vague_definition** BPS or a key construct is used without a definition.
- **tokenistic_bps** BPS appears as a label with no analytic follow-through.
- **missing_social** Social domain absent despite a BPS claim.
- **missing_biology** Biological domain absent despite a BPS claim.
- **mechanistic_absence** No cross-domain mechanism is offered.
- **construct_overlap** Constructs are used interchangeably or with blurred boundaries.
- **parallel_listing_without_integration** Domains are listed side by side with no integration.
- **none** No inferable conceptual problem.

provisional_typology

Abstract-level BPS signal, refined later at Stage 3.

- **potential integrative signal** All three domains substantively present with a cross-domain signal.
- **multifactorial signal** All three domains present but mainly parallel.
- **pseudo-bps or partial signal** One or more core domains thin or absent despite BPS language.
- **rhetoical label signal** BPS mainly symbolic or confined to framing or conclusion.

stage3_candidate / stage3_priority

Whether and how urgently the record proceeds to Stage 3.

- **yes / no** Candidate flag.
- **high / medium / low** Retrieval and coding priority.

coding_rationale

One-sentence rationale supporting the coding bundle. (*free text*)

coding_method / llm_model

Operational provenance of the row (llm_structured, llm_batch_fallback, rule_based_fallback) and the model identifier when the LLM stage was used.

Prompt Rules Used in the Structured LLM Stage

These constraints are embedded in the structured prompt and are the operational meaning of the codes above.

- Code only from title, abstract, publication types, and journal metadata. Do not infer content that is absent from the record.
- Do not treat the single lexical token biopsychosocial as proof that all three domains are substantively present.
- Use explanatory framework only when BPS explicitly explains pain or pain-related disability.
- Use intervention rationale when BPS mainly justifies multimodal or interdisciplinary treatment.
- Use organizing principle when BPS structures scope or categories without integration mechanisms.
- Use rhetorical label when BPS is ceremonial, aspirational, or symbolic with no substantive analytic work.
- Set stage3_candidate to yes for musculoskeletal or neuropathic reviews, and for mixed or unspecified chronic pain reviews where either relevance cannot be ruled out. The two planned reviews (musculoskeletal and neuropathic) draw from this same candidate pool, routed by pain-condition family.
- Use potential integrative signal only when all three core domains are substantively present and the abstract signals cross-domain explanation or organization beyond simple listing.
- Return conceptual_problem_flags as none only when no inferable conceptual problem is present.

Deterministic Repair and Fallback Logic

How the pipeline guarantees a complete, in-vocabulary row.

- Missing or malformed model responses are repaired against the fixed vocabularies above.
- Aliases such as scoping review or narrative review are normalized to the implemented labels.
- stage3_candidate is forced to yes whenever musculoskeletal_flag or neuropathic_flag is yes or unclear, and stage3_track is set from whichever flags fire (musculoskeletal, neuropathic, or both).
- Conceptual problem flags are post-derived from typology, BPS function, and missing domains: rhetorical usage triggers tokenistic_bps and vague_definition; parallel non-mechanistic usage triggers parallel_listing_without_integration and mechanistic_absence; absent biological or social content triggers missing_biology or missing_social.
- When the LLM stage fails, the rule-based fallback still generates review type, objective category, ICD-11 category, BPS function, domain mention flags, concept detections, and provisional Stage 3 priority.

Proposed Refinement: Graded Domain Mention and Dual Function

PROPOSED REFINEMENT

Awaiting expert evaluation. Not yet applied to the pipeline.

Refinement proposed for expert evaluation. Two changes would raise the resolution of Stage 2 without changing its scope.

First, replace the binary `bio_mentioned`, `psych_mentioned`, and `social_mentioned` with a three-level scale: substantive (a named construct is discussed), lexical (only the domain umbrella word appears), and absent. This mirrors the `S_lex` versus `S_subst` distinction already used in the manuscript audit and lets the social-shortfall finding be measured at the abstract level rather than only after recoding.

Second, allow `bps_function` to carry a dominant function plus an optional secondary function. Many abstracts both organize scope and justify treatment; forcing a single label discards that signal and depresses agreement on mixed abstracts.

Third, reinstate optional `spiritual_existential` and `lifestyle_mention` flags. The corpus contains explicit biopsychosocial-spiritual reviews and religious-belief reviews, so the extra flags capture real variance that the current binary domains drop.

Worked Examples (real corpus records)

Actual Stage 2 codings from the test-run manifest, shown as reliability anchors.

Where has the 'bio' in bio-psycho-social gone?

Narrative or expert review, 2019

Coding: `provisional_typology` = pseudo-bps or partial signal; `conceptual_problem_flags` = parallel_listing_without_integration; `bps_function` = organizing principle.

Why: The title itself signals a missing biological pole, so the typology is a partial signal and the flags record the imbalance.

The biopsychosocial model is lost in translation: from misrepresentation to an enactive modernization

Narrative or expert review, 2023

Coding: `provisional_typology` = potential integrative signal; `conceptual_problem_flags` = vague_definition | tokenistic_bps; `bps_function` = explanatory framework.

Why: A conceptual, theory-facing review can be an integrative signal and still be flagged for definitional vagueness. The two axes are independent.

Exercise for chronic musculoskeletal pain: a biopsychosocial approach

Narrative or expert review, 2017

Coding: `icd11_pain_category` = chronic secondary musculoskeletal pain; `stage3_candidate` = yes; `provisional_typology` = potential integrative signal.

Why: Musculoskeletal focus forces the Stage 3 candidate flag; the integrative signal sets a high retrieval priority.

Documented Divergences

Where prose, CSV codebook, and outputs disagree, the output schema is treated as canonical.

- The prose codebook lists treatment-oriented as an objective class, but the implemented prompt and output schema do not use that label.
- The CSV codebook historically contains `spiritual_existential_mentioned` and `lifestyle_mentioned`; those fields are absent from the current output file and are therefore not canonical here. The refinement above proposes reinstating them explicitly.
- Prose and code both emphasize that BPS language alone is insufficient evidence of true triadic coverage.

- The test-run implementation only carried `musculoskeletal_flag`. This revised scheme adds `neuropathic_flag` and `stage3_track` as first-class fields so the musculoskeletal and neuropathic reviews are seeded from the same Stage 2 pass with one shared vocabulary. These fields are specified here for expert review and will be wired into the code on the post-sign-off re-run.

Primary Outputs Using This Scheme

- `src/review_stages/04_extraction/outputs/stage2_abstract_coding.csv`
- `src/review_stages/04_extraction/forms/stage2_double_code_subset.csv`
- `src/review_stages/04_extraction/outputs/stage2_objective_llm_assist.csv`
- `src/review_stages/04_extraction/outputs/llm_objective_pilot.json`